Use-Case Flow Specification

Personalized Meal & Diet Subscription Manager — UC-02: Generate Weekly Meal Plan

Primary Actor	Subscriber
Supporting Actor	Dietitian

1. Preconditions

1. The Subscriber has an active meal subscription.
2. The Subscriber's dietary profile is available to the system.
3. The profile contains the applicable dietary constraints, including allergies, diabetic limits, and calorie goals.
4. The Dietitian is authorized to review the generated meal plan.

2. Main Success Scenario

1. The Subscriber requests generation of the weekly meal plan.
2. The system retrieves the Subscriber's dietary profile.
3. The system retrieves the applicable allergies, diabetic limits, calorie goals, and other dietary constraints.
4. The system validates the dietary constraints.
5. The system generates a weekly meal plan that adheres to the validated constraints.
6. The system checks the generated menu against allergen exclusions and nutritional limits.
7. The system stores the generated meal plan.
8. The system makes the generated meal plan available to the Dietitian for review.
9. The Dietitian reviews and approves the meal plan.
10. The system makes the approved weekly meal plan available to the Subscriber.

3. Alternate Flow — Dietary Constraint Violation

5a. The generated meal plan violates an allergen exclusion or another dietary constraint.

5a1. The system rejects the generated meal plan.

5a2. The system identifies the violated constraint.

5a3. The system regenerates the meal plan using the valid constraints.

5a4. The system rechecks the regenerated meal plan.

5a5. If all constraints are satisfied, the process continues at Step 7.

4. Postconditions

1. A weekly meal plan satisfying the Subscriber's dietary constraints is stored in the system.
2. The approved meal plan is available to the Subscriber.
3. The meal plan can be used for the Subscriber's scheduled meal subscription.